

Effects of a 6-Week Repeated-Sprint Training With Voluntary Hypoventilation at Low and High Lung Volume on Repeated-Sprint Ability in Female Soccer Players

Mounir Ait Ali Braham,¹ Youva Ouchen,² and Xavier Woorons^{3,4}

¹Département des Sciences de l'Activité Physique Trois-Rivières, Université du Québec à Trois-Rivières, Trois-Rivières, QC, Canada; ²Département STAPS, UFR SMBH, Université Sorbonne Paris Nord, Bobigny, France; ³EA 7369-URPSSS-Unité de Recherche Pluridisciplinaire Sport Santé Société, Université Lille, Université Artois, Université Littoral Côte d'Opale, Lille, France; ⁴ARPEH, Association for Research and Promotion of Hypoventilation Training, Lille, France

Purpose: To investigate the effects of repeated-sprint training with voluntary hypoventilation at low (RSH-VHL) and high (RS-VHH) lung volume on repeated-sprint ability (RSA) in female athletes. **Methods:** Over a 6-week period, 24 female soccer players completed 12 sessions of repeated 30-m running sprints with end-expiratory breath holding (RSH-VHL, $n = 8$), end-inspiratory breath holding (RS-VHH, $n=8$), or unrestricted breathing (RS-URB, $n=8$). Before and after training, a running RSA test consisting of performing 30-m all-out sprints until exhaustion was implemented. **Results:** From before to after training, the number of sprints completed during the RSA test was increased in both RSH-VHL (19.3 [0.9] vs 22.6 [0.9]; $P<.01$) and RS-VHH (19.3 [1.5] vs 20.5 [1.7]; $P<.01$) but not in RS-URB (19.4 [1.3] vs 19.5 [1.7]; $P=.67$). The mean velocity and the percentage decrement score calculated over sprints 1 to 17 were, respectively, higher (82.2% [1.8%] vs 84.6% [2.1%] of maximal velocity) and lower (23.7% [3.1%] vs 19.4% [3.2%]) in RSH-VHL ($P<.01$), whereas they remained unchanged in RS-VHH and RS-URB. The mean arterial oxygen saturation recorded during training at the end of the sprints was lower in RSH-VHL (92.1% [0.4%]) than in RS-VHH (97.3% [0.1%]) and RS-URB (97.8% [0.1%]). **Conclusions:** This study shows that female athletes can benefit from the RSH-VHL intervention to improve RSA. The performance gains may have been limited by the short sprinting distance with end-expiratory breath holding, which provoked only moderate hypoxemia. The increase in the number of sprints in RS-VHH seems to show that factors other than hypoxia may have played a role in RSA improvement.

Keywords: hypoxia, hypercapnia, breath holding, VHL, RSH

Soccer is a sport involving intermittent high-intensity efforts that include short sprints, accelerations, rapid decelerations, changes of direction, and jumps.¹ The ability to recover rapidly from these efforts in order to maintain a high level of intensity throughout a soccer match is paramount. In particular, the ability to repeat sprints, namely the repeated-sprint ability (RSA), is an important fitness requirement in soccer which may influence the final outcome of the match.²

As a multifactorial fitness component, RSA can be improved using different training methods.³ However, over the past 10 years, it has been shown that one of the best ways to improve RSA was to perform a repeated-sprint training under hypoxic conditions (RSH). Most of

the studies dealing with this approach have reported larger performance gains than the same training performed in normoxic conditions.^{4,5} However, because the access to sufficiently high altitude is difficult in many countries and considering that the use of hypoxia needs a specific logistic and a high cost, RSH is not accessible to many athletes.

Since the mid-2000s, it has been shown that it could be possible to exercise in hypoxia while remaining in a normoxic environment when using a voluntary hypoventilation at low lung volume (VHL).⁶⁻⁸ This method, which consists in exercising while performing short bouts of end-expiratory breath holding, has been reported to induce a fall in arterial oxygen saturation leading to an accentuated

Ait Ali Braham

<https://orcid.org/0009-0000-8573-7500>

Woorons (xavier.woorons@gmail.com) is corresponding author, <https://orcid.org/0000-0002-8150-9153>

muscle deoxygenation.^{6,9} After several weeks of RSH induced by VHL (RSH-VHL), some studies have reported larger improvements in RSA, as compared to the same training performed in normoxia. In open-loop RSA test designs, highly trained swimmers¹⁰ and rugby players' improved the maximum number of sprints performed until exhaustion after 2 and 4 weeks of RSH-VHL. In closed-loop RSA tests, the mean power output and the mean velocity were also increased in cyclists¹² and ice hockey players,¹³ respectively, after 3 and 5 weeks of such training regimen. Furthermore, in the same RSA test design performed in running, a reduction in fatigue has been reported in basketball players after 4 weeks of running RSH-VHL¹⁴ and in team sport players after 3 weeks of supramaximal cycling exercises with VHL.¹⁵

So far, it is noteworthy that the studies that have investigated the effects of RSH-VHL on RSA have been conducted either exclusively or mostly in men. The effectiveness of this training method has therefore to be confirmed in women, in particular because some of their physiological responses to exercise and training might differ from those of their male counterparts.¹⁶⁻¹⁸ Furthermore, to the best of our knowledge, no study has ever compared the effects of RSH-VHL versus a repeated-sprint training with voluntary hypoventilation at high lung volume (RS-VHH). While exercising with VHH has not been reported to elicit any hypoxic effect, it may provoke hypercapnia under certain conditions.^{19,20} The comparison of RSH-VHL with RS-VHH could therefore help establish whether factors other than hypoxia may lead to RSA improvement, if any.

Regarding the considerations mentioned above, the goal of the present study was to determine the effects of a 6-week period of RSH-VHL and RS-VHH on RSA performance in youth female

soccer players. We hypothesized that both training conditions would improve performance as compared to the same training performed with unrestricted breathing but with a superiority of RSH-VHL over RS-VHH.

Subjects

Methods

Twenty-four youth female soccer players participated in this study (Table 1). All subjects were eumenorrheic women with normal regular menstrual cycles. The study was approved by the local Ethics Committee (2021-483-S92) and complied with the Declaration of Helsinki (2008). All subjects and their parents (for minors) were informed about the risks, benefits, and procedures of the study before providing their written informed consent prior to participation.

Study Design

The experiments were conducted during the precompetitive period and the start of the competitive season (from mid-August to the end of September). The protocol consisted in performing one RSA testing session before and after 6 weeks of repeated-sprint training in running (2 sessions per week). Testing and training sessions were performed at sea level and on a synthetic outdoor football field. The subjects were matched into groups of 3 according to their performance in the RSA test and then randomly assigned to the group that performed the repeated-sprint training either with unrestricted breathing (RS-URB), with VHL (RSH-VHL), or with VHH (RS-VHH). Before starting the experiment, 2 sessions were implemented to familiarize the participants with the VHL and the VHH procedures. The VHL technique was well described previously.²¹ Briefly, the subjects had to exhale down to around the functional residual capacity just before the start of the sprints and then had to hold their breath until the end of the sprint. For the VHH procedure, the subjects were required to perform each sprint while holding breath at the end of inhalation. In this regard, subjects were required not to perform a forced inhalation but to hold

breath at the end of a normal inhalation.

Training Protocol

Over a 6-week training period, subjects had to

complete 12 sessions of 30-m straight line running repeated sprints. Subjects were asked to perform the sprints at maximal velocity and to respect the breathing modality of the group they were assigned to. All the training sessions were fully supervised, and we carefully checked visually that the subjects were applying the VHL and VHH techniques correctly

Table 1 Characteristics of the Female Soccer Players Who Participated in the Study

	RS-H-VHL	RS-VHH	RS-URB
U17 team, n	4	4	3
U19 team, n	4	4	5
Age, y	16.3 (0.6)	16.4 (0.7)	
	169.3 (3.5)		
	62.6 (4.1)	171.1 (4.7) 62.8 (6.0)	
Height, cm			16.8 (0.7) 174.0
Weight, kg			(3.2) 65.3 (4.9)

Abbreviations: RS-URB, repeated-sprint training with unrestricted breathing; RS-VHH, repeated-sprint training with voluntary hypoventilation at high lung volume; RS-H-VHL, repeated-sprint training in hypoxia induced by voluntary hypoventilation at low lung volume. Note: Data for age, height, and weight are mean (SD).

at each repetition. Each repeated-sprint session was preceded by a 15-minute warm-up (active mobilization, dynamic stretching, running drills, and progressive accelerations). During the first week of the training period, subjects had to perform 2 sets of 8 x 30 m. Then, 1 repetition per set was added in each following week. In the last week, to avoid performing >12 repetitions per set, subjects had to complete 3 sets of 9 x 30 m. For all repeated-sprint sessions, and in all groups, a 25-second semiactive recovery period (ie, walking with unrestricted breathing) was observed after each repetition, whereas sets were separated by 5 minutes of passive recovery.

Testing Protocol

Four days before (Pre) and 4 days after (Post) the training period, a running RSA test until exhaustion (voluntary cessation) was implemented. Prior to testing, the same warm-up as during the repeated-sprint sessions was

completed. The RSA test consisted of the repetition of straight line 30-m running sprints each performed at maximal velocity of the RSA test (V_{max}) and with unrestricted breathing. Each sprint was followed by a 25-second semiactive recovery. A countdown was given 5 seconds before the start in order to allow the subjects to assume the standing ready position. Participants were given strong verbal encouragements during the entire test to complete as many sprints as possible.

Measurements

Testing Data

Time of each sprint of the RSA test was measured with photocells (Witty, Microgate). The total number of sprints completed until voluntary cessation was evaluated, and the rating of perceived

of the test. V_{max} was calculated from the time of the fastest sprint of exertion (RPE, 10-point Borg scale) was assessed just at the end of the test. Velocity of all subsequent sprints was also calculated and expressed as percentage of V_{max} . We chose to also evaluate RSA performance over a fixed number of sprints (closed-loop approach analysis) which corresponded to the greatest number of sprints that could be performed by all the subjects of the 3 groups both at Pre and Post (defined as x). For this purpose, we calculated the mean velocity (as percentage of V_{max}) from sprint 1 to sprint x ($V_{mean[1-x]}$). In addition, according to the same closed-loop approach, fatigue was assessed through the percentage decrement score as follows:

$$[(\text{total sprint time} / \text{ideal sprint time}) - 1] \times 100,$$

where total sprint time is the sum of sprint 1 to sprint x , and ideal sprint time is the product of the time of the fastest sprint and x . This formula has been found to be the most valid and reliable method to estimate RSA-induced fatigability.²²

Training Data

In the third week of the training period, just before the start of the competitive season, measurements were made during a repeated-sprint session including 2 sets of 10 x 30 m to assess and compare the effects of the 3 conditions on performance (time of sprints), arterial oxygen saturation (SpO_2 , Nonin Onyx II) and RPE.

Statistical Analysis

Before performing the analyses, distribution normality and variance homogeneity of data were tested with Shapiro-Wilk and Levene tests, respectively. Two-way analyses of variance (ANOVA) for repeated measures were used to determine whether there was an

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effect of training condition (RSH-VHL vs RS-VHH vs RS-URB) and/or time (Pre vs Post) in the mean of the variables measured during the RSA test. We also used 2-way repeated-measures ANOVA to analyze the change in velocity for each sprint of the RSA test between Pre and Post within training conditions and between training conditions both at Pre and Post. The analyses of training data were also performed through 2-way repeated-measures ANOVA. When a significant effect was found, the Bonferroni post hoc procedure was used to localize the difference. All data analyses were conducted with the software Sigmapstat (version 3.5, Systat Software). Data are presented as mean (SD). Significance level was set at $P < .05$.

Training Data

Results

In the 3 groups, all subjects completed the 12 training sessions and 257 sprints in total. Data recorded during the third week of the training period show that there was no difference between groups in mean time of the 30-m sprints both in set 1 (S1) and set 2 (S2; Figure 1A). The mean time of sprints was higher in S2 than in S1 in RSH-VHL and RS-URB ($P < .01$) but not

RSH-VHL in Youth Female Soccer Players 3

different in RS-VHH ($P = .41$). The mean SpO₂ recorded at the end of the sprints (ANOVA P values: training condition [C] $< .01$, time [T] $.03$, interaction [I] $< .04$) was lower in RSH-VHL (S1: 92.5% [0.9%]; S2: 92.0% [1.0%]) than in both RS-VHH (S1: 97.1% [0.5%]; S2: 97.0% [0.7%]) and RS-URB (S1: 97.4% [0.6%]; S2: 97.4% [0.7%]; $P < .01$) and not different between RS-VHH and RS-URB ($P > .60$) in the 2 sets. SpO₂ was also lower in RSH-VHL than in the 2 other conditions during each sprint of S1 and S2 (Figure 1B), whereas there was no difference between RS-VHH and RS-URB. The mean RPE of the training session was higher in RSH-VHL (8.1 [0.8]) than in both RS-VHH (7.2 [0.5]; $P = .03$) and RS-URB (7.2 [0.5]; $P = .02$) and not different between RS-VHH and RS-URB ($P = .50$).

Testing Data

The number of sprints completed until exhaustion during the RSA test was

greater at Post than at Pre in RSH-VHL and in RS-VHH ($P<.01$) but remained unchanged in RS-URB ($P=.67$; Table 2). Furthermore, while the number of sprints was not different between groups at Pre ($P>.90$), it was higher in RSH-VHL than in both

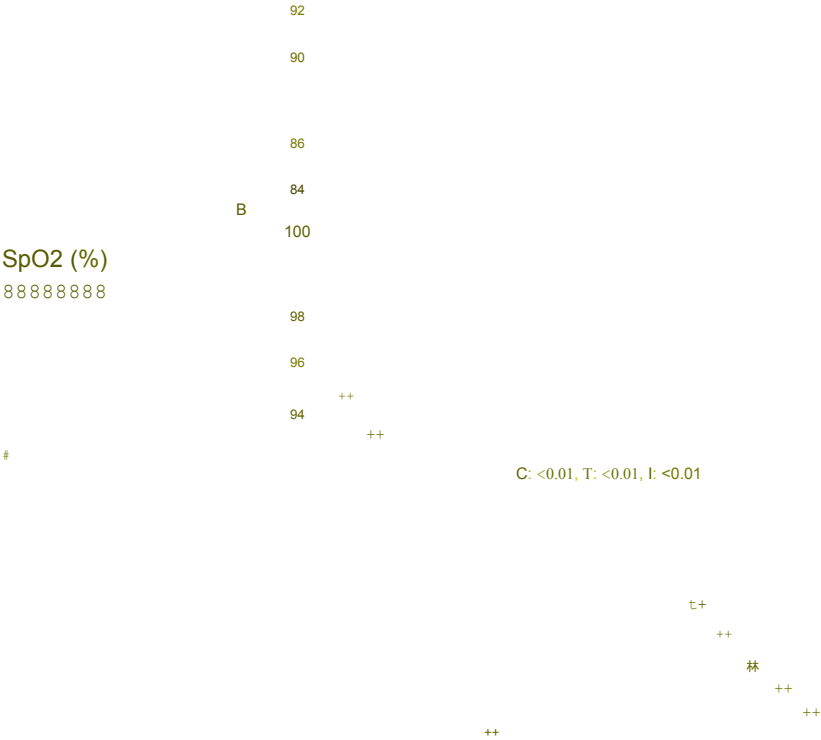
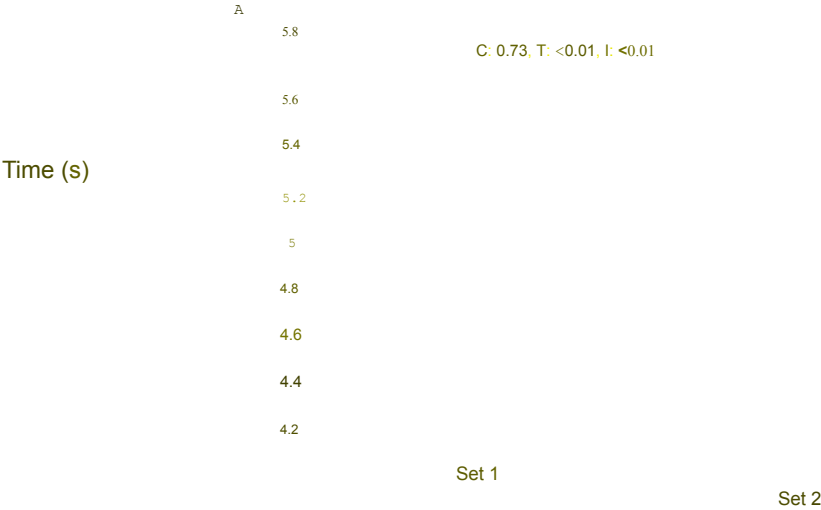




Figure 1 (A) Mean time of the 30-m sprints in set 1 and set 2 of an RSH-VHL (white bars), RS-VHH (black-and-white bars), and RS-URB (black bars) session. C indicates training condition effect; T, time effect (set 1 vs set 2); I, interaction effect; RS-URB, repeated-sprint training performed with unrestricted breathing; RS-VHH, repeated-sprint training performed with voluntary hypoventilation at high lung volume; RSH-VHL, repeated-sprint training performed with voluntary hypoventilation at low lung volume; SpO₂, arterial oxygen saturation. **P* .05 for difference with set 1. (B) Mean SpO₂ recorded at the end of each sprint of set 1 and set 2 during an RSH-VHL (white circles), RS-VHH (white squares), or RS-URB (black circles). ‡*P* < .05 for difference with RS-VHH. †*P* < .05 for difference with RS-URB.

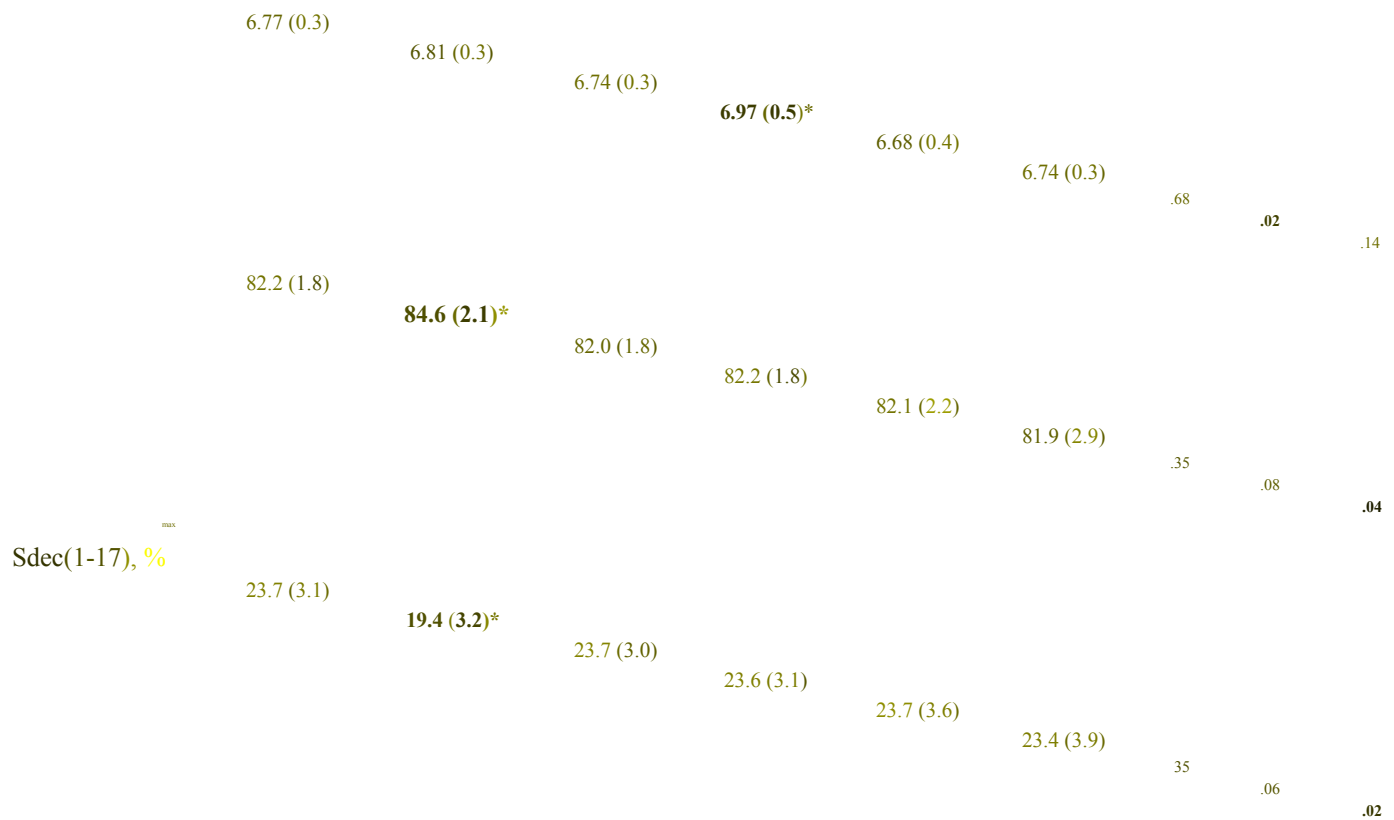
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Table 2 Results of the RSA Test

	RSH-VHL		RS-VHH		RS-URB		ANOVA P values		
	Pre	Post	Pre	Post	Pre	Post	C	T	I
Number of sprints	19.3 (0.9)	22.6 (0.9)*‡†	19.3 (1.5)	20.5 (1.7)*	19.4 (1.3)	19.5 (1.7)	.08	<.01	<.01
RPE	9.5 (0.5)	9.6 (0.5)	9.6 (0.4)	9.7 (0.4)	9.4 (0.5)	9.4 (0.5)	.53	.58	.91
V _{max} , m·s ⁻¹	1.7 (0.1)	1.7 (0.1)	1.7 (0.1)	1.7 (0.1)	1.7 (0.1)	1.7 (0.1)	.91	.91	.91
V _{mean} (1-17), m·s ⁻¹	1.6 (0.1)	1.6 (0.1)	1.6 (0.1)	1.6 (0.1)	1.6 (0.1)	1.6 (0.1)	.91	.91	.91
%V _m	100 (0)	100 (0)	100 (0)	100 (0)	100 (0)	100 (0)	.91	.91	.91



Abbreviations: ANOVA, analysis of variance; RPE, rating of perceived exertion at the end of the RSA test; RSA, repeated-sprint ability; RS-**URB**, repeated-sprint training **with** unrestricted breathing; RSH-VHL, repeated-sprint training in hypoxia induced **by** voluntary hypoventilation at low lung volume; RS-VHH, repeated-sprint training with voluntary hypoventilation at high lung volume; Sdec(1-17), percentage decrement score over sprints 1 to 17; Vmax, maximal velocity; Vmean(1-17), mean velocity over sprints 1 to 17. Note: Values are mean (SD). Bold values indicate significant differences.

$P < .05$ for difference with VHH. $\dagger P < .05$ for difference with URB. $*P < .05$ for difference with Pre within groups.

RS-VHH ($P = .01$) and RS-URB ($P < .01$) at Post and not different between RS-VHH and RS-URB ($P = .47$). There was no difference in RPE at the end of the RSA test between Pre and Post within groups and between groups both at Pre and Post. Data for each of the subjects in the 3 groups are shown in Figure 2.

The Vmax reached during the RSA test was not different between Pre and Post in RSH-VHL ($P = .67$) and RS-URB ($P = .42$) and was higher at Post in RS-VHH ($P < .01$). There was no difference in Vmax between training conditions.

The greatest number of sprints that could be performed by all the subjects of the 3 groups both at Pre and Post was 17. The closed-loop approach analysis shows that the mean velocity over sprint 1 to 17 (Vmean[1-17]) was higher at Post than at Pre in RSH-VHL ($P < .01$) but did not change in RS-VHH ($P = .83$) and RS-URB ($P = .80$; Table 2; individual data are presented in Figure 2). There was no difference in Vmean(1-17) between conditions both at Pre and Post.

The percentage decrement score over sprint 1 to 17 (Sdec[1-17]) was lower at Post than at Pre in RSH-VHL ($P < .01$) and remained unchanged in RS-VHH ($P = .88$) and RS-URB ($P = .73$). In

addition, while $S_{dec}(1-17)$ was not different between conditions at Pre, it tended to be lower in RSH-VHL than in RS-URB at Post ($P=.06$) and was not different between RSH-VHL and RS-VHH ($P=.12$) and between RS-VHH and RS-URB ($P=.98$).

The sprint-by-sprint analysis (over sprint 1 to 17) shows that velocity was higher at Post than at Pre from sprint 8 in RSH-VHL ($P<.01$), whereas there was no change in RS-VHH ($P = .77$) and in RS-URB ($P = .82$; Figure 3). Furthermore, while velocity was not different between conditions at Pre whatever the sprint number, it was higher in RSH-VHL than in RS-VHH at sprints 11, 12, and 13 and higher in RSH-VHL than in RS-URB at sprints 16 and 17.

Discussion

2 weeks and 6 sessions of this training intervention, highly trained swimmers (mostly male) significantly improved the number of sprints until exhaustion in an open-loop RSA test (+35%).¹⁰ With the same testing design, rugby players of national level dramatically increased the maximal number of sprints after 7 sessions (4 wk) of RSH-VHL (+64%). Other studies have reported 8% and 4% performance gains in RSA tests with a fixed number of sprints (closed-loop testing) in a cohort of male cyclists¹² and ice hockey players,¹³ respectively, after 3 and 5 weeks of RSH-VHL. In the present study, the 17% increase in the maximal number of sprints completed during the RSA test and the 2.4% improvement of

demonstrate that female athletes can also benefit from $\text{mean}(1-17)$ RSH-VHL. However, these gains seem to be of a lesser extent than what has been previously reported in their male counterparts.

Several reasons may explain a moderate rather than a large RSA improvement in the female soccer players after RSH-VHL.

First, and probably most importantly, the 30-m distance used for the repeated-sprint sessions was probably too short to provoke a marked drop in SpO_2 when performed with end-expiratory breath holding. Under this condition, the mean sprinting time was 5 seconds with mean minimal and average SpO_2 of 90% and 92%, respectively. In the previous RSH-VHL studies, which used 40-m running sprints (breath hold time of about 6.3 s),^{11,13} or 6-second sprint duration,^{12,13} these values were, respectively, of 83%^{10,11,13} and 88%.^{12,13} It is likely that the mild hypoxemia recorded in the present study reduced the hypoxic stimulus and the subsequent physiological adaptations with, ultimately, a lower positive impact on RSA performance. Second, with regard to the demanding feature of the RSH-VHL method, the duration of the training period, which was longer than in the previous studies, may have led to fatigue and attenuated the RSA improvement at Post.¹³ Third, in relation to the length of the training period, the timing of the Post testing session may have also played a role. In the studies of Fornasier Santos et al¹¹ and Brocherie et al,¹³ that were also carried out in running, the post-RSA test was implemented 1 week after the training period whose duration was 4 and 5 weeks, respectively. In the present study, one may hypothesize that the

posttesting session was placed too soon after the last training session (ie, 4 d) to allow subjects to recover optimally from the 6-week repeated-sprint training period and therefore to obtain higher RSA gains. Fourth, it is possible that the menstrual cycle status of the female participants influenced the results. Indeed, the variations of sex hormone concentrations during the phases of the menstrual cycle may alter force production and therefore muscle strength and power.²³ In female handball players performing repeated cycle sprint exercises, a decrease in peak power output, followed by a reduction in

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To the best of our knowledge, this study is the first to investigate the effects of RSH-VHL on RSA in female athletes. It is also the first to compare RSH-VHL with repeated-sprint training with VHH. The results show that female soccer players of the RSH-VHL group significantly improved RSA after 6 weeks of training, whereas performance remained unchanged in the RS-URB group. The results also reveal an improvement in the number of sprints completed until exhaustion in the RS-VHH group, but which was of a lower extent than in RSH-VHL.

The interest of using RSH-VHL for improving RSA in male athletes has been demonstrated in several previous studies. After

Number of sprints
Number of sprints
26
25
24
23
22
21
20
19
18
17
16
15

Pre

Post

RS-VHH
RSH-VHL

Vmean(1-17) (%Vmax)

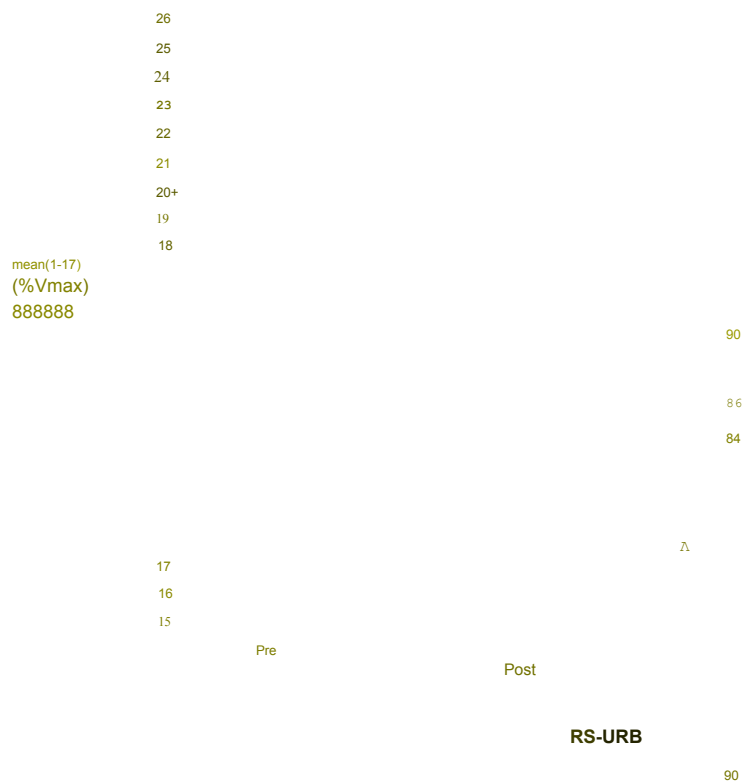
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90
86
84
82
80
78
76
74

RSH-VHL in Youth Female Soccer Players

Pre

Post



2

2

2

2

2

2

2

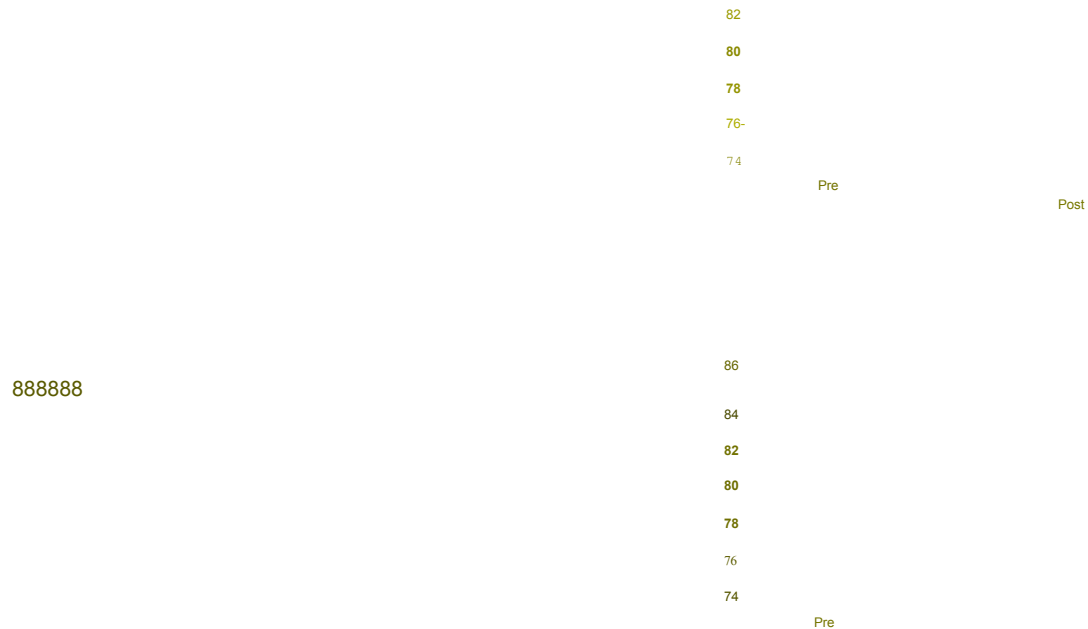


Figure 2 - Number of sprints completed until exhaustion during the repeated-sprint-ability test (left panels) and $V_{mean}[1-17]$ (expressed as percentage of maximal velocity; right panels) for each individual subject Pre and Post RSH-VHL (top panels), RS-VHH (middle panels), and RS-URB (bottom panels). Post indicates after; Pre, before; RS-URB, repeated-sprint training performed with unrestricted breathing; RS-VHH, repeated-sprint training performed with voluntary hypoventilation at high lung volume; RSH-VHL, repeated-sprint training in hypoxia performed with voluntary hypoventilation at low lung volume; $V_{mean}(1-17)$, mean velocity over sprints 1 to 17.

maximal voluntary contraction of the knee extensor, has been reported during the late luteal phase of the menstrual cycle as compared to the follicular phase.²⁴ While it cannot be excluded that this issue (slightly) reduced the effectiveness of the RSH-VHL intervention, it may also have impacted RSA in the Pre and Post testing sessions. Finally, the young age of the participants may also have had an impact on RSA performance regardless of the training conditions.

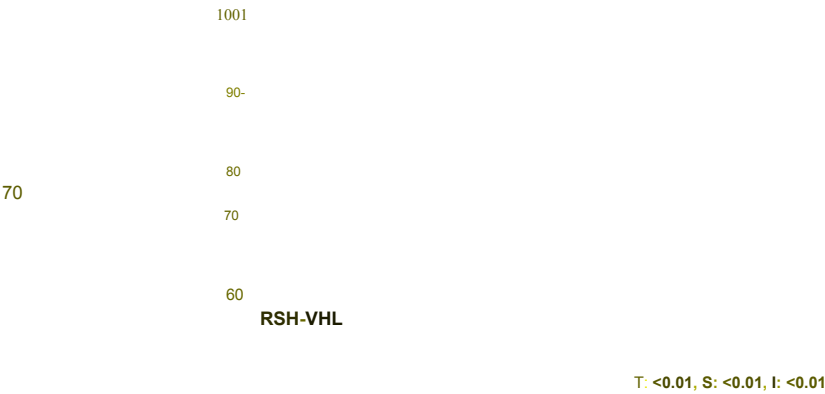
oxygen uptake¹² and enhanced muscle reoxygenation¹³ have been found during a RSA test after 6 and 8 sessions of RSH-VHL.

Such physiological adaptations are paramount for repeated-sprint exercise since they increase the oxygen availability to the muscles, which improves phosphocreatine resynthesis and the removal of waste metabolites during the recovery periods following sprints.²⁵ Considering that during repeated-sprint exercise the contribution of the aerobic metabolism becomes predominant after completing several sprints,²⁵ the higher velocity recorded from the eighth sprint in the female soccer players of the RSH-VHL group is in accordance with the hypothesis of an improved oxygen availability to the muscles in these subjects. Lower blood potassium ion concentrations correlated to enhanced RSA performance have also been found after 4 weeks of RSH-VHL in basketball players.¹³ These changes may result from an improvement in muscle contractile function,²⁶ which could lead to lower muscle fatigue during intense exercises. We could speculate that these physiological adaptations

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When analyzing the sprint-by-sprint performance throughout the RSA test, it is noticeable that the increase in running velocity after the RSH-VHL intervention occurred only from the eighth sprint. This outcome is quite similar to what has been reported in the previous studies in which performance was improved in the second part (or mainly) of the RSA test after RSH-VHL.¹⁰⁻¹⁴ Therefore, it is probable that the female soccer players of the present study had the same physiological adaptations as those previously reported in male athletes. In some studies, higher

Velocity (%Vmax)



50

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23

RS-VHH



Velocity (%Vmax)

898

Velocity (%Vmax)

50

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23

RS-URB

100



Figure 3

Mean velocity (expressed as percentage of maximal velocity) for each repetition of the repeated-sprint-ability test Pre and Post RSH-VHL, RS-VHH, and RS-URB. T indicates time effect (Pre vs Post); S, sprint-number effect; I, interaction effect; Post, after; Pre, before; RS-URB, repeated- sprint training performed with unrestricted breathing; RS-VHH, repeated-sprint training performed with voluntary hypoventilation at high lung volume; RSH-VHL, repeated-sprint training in hypoxia performed with voluntary hypoventilation at low lung volume. **P* .05 for difference with sprint of same number in Pre in the RSH-VHL group. ‡*P* < .05 for difference with sprint of same number in RS-VHH. †*P* < .05 for difference with sprint of same number in RS-URB. N.B.: Statistical analyses were performed from sprints 1 to sprint 17.

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also occurred in this study and may have played a role in the RSA gain after RSH-VHL. Finally, high-intensity VHL training and RSH-VHL have been shown to markedly increase the glycolysis activity, in particular through large improvement in maximal blood lactate concentration during specific anaerobic trials. 10,12,27 However, after several weeks of RSH-VHL, no change in maximal blood lactate concentration and no difference in this parameter with the control group have been found during an RSA test. 11-13 Therefore, it is unlikely that a greater anaerobic capacity or higher lactate tolerance was involved in the RSA gains recorded in the

female soccer players.

Another original goal of this study was to determine the effects of the RS-VHH approach on RSA. Our results show that the female soccer players who followed this training intervention increased the maximal number of sprints until exhaustion by 6.2%. On the other hand, they did not improve $V_{mean}(1-17)$ or $S_{dec}(1-17)$. It has been reported that exercising with breath holding at high lung volume produces hypercapnia under certain conditions.^{28,29} However, it is unlikely that 5-second sprints (on average) with the lungs full of air could have induced a significant hypercapnic effect. Therefore, it is probable that factors other than hypoxia and hypercapnia, possibly psychological (see Limitations section), played a role in the augmented number of sprints in the RS-VHH group.

The first limitation of the present study is that no physiological parameters were measured during the testing sessions. Therefore, it was not possible to show the mechanisms that may have improved RSA in particular in the RSH-VHL group. However, the physiological adaptations induced by this approach have been specifically investigated previously.^{12,14} Despite a lower hypoxic stimulus, it is likely that the physiological mechanisms that led to RSA improvement in the RSH-VHL group were the same, although possibly of lower extent, as those presented above. Still our study cannot really explain why the RSH-VHL approach was superior to the RS-VHH one. Another limitation is that the menstrual cycle status and the hormonal variations of the female participants were not controlled. The influence of these factors on both the training and testing sessions could therefore not be assessed. Yet, the present work was not designed to observe changes throughout the menstrual cycle, which should be investigated in future studies. A third point concerns the fact that, like in all VHL/VHH studies, subjects, as well as investigators, could not be blinded. Therefore, a placebo effect cannot be excluded, especially for the maximum number of sprints performed until voluntary cessation. On the other hand, this is more improbable for the closed-loop approach, in particular because subjects (and investigators) did not know in advance the number of sprints that would be taken into account. Furthermore, the role of a placebo effect to explain the RSA gains after RSH-VHL has been recently challenged.¹⁵

Practical Applications

The effectiveness of RSH to improve RSA in different sporting disciplines has been well established.^{4,5} Female soccer players who could not train at high (simulated) altitudes (ie, above 2500 m) to implement RSH could advantageously use the end-expiratory (ie, VHL) rather than end-inspiratory (VHH) breath hold technique. The fact that the performance gains after RSH-VHL occur especially in the second part of an RSA test may provide a physical advantage in the final part of a football game. It can for instance give the possibility to win possession of the ball and

increase the chances of scoring while preventing the opponents to do so.²

Conclusions

This study showed that 12 sessions of repeated-sprint training with voluntary hypoventilation at low lung volume improved repeated-sprint ability in youth female soccer players, thus demonstrating that women can also benefit from this kind of intervention. The moderate hypoxic effect resulting from the short sprinting distance with end-expiratory breath holding may have limited the extent of the performance gains. The increase in the maximal number of sprints after the repeated-sprint-training with end-inspiratory-breath-holding intervention seems to show that factors other than hypoxia may have played a role in the repeated-sprint ability improvement.

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